Final Examination Practice Sheet

Course Code: CSE563
Course Title: APPLIED MATHEMATICS
Section(s): K3P25RB
Syllabus Covered: Unit 1 to Unit 6
Pattern: per unit, 8 MCQs (1 mark) and 2 long questions (10 marks)
Purpose: self-practice for the end-term examination

Instructions: attempt every question without notes, then check your answers against the key. For the long questions, write structured answers in the form you would use in the examination, and show all working for numerical parts.

Unit 1: Basic Mathematics in Computer Science

Part A - Multiple Choice Questions (8 x 1 = 8 Marks)

1. What is the greatest common divisor of 48 and 180?
(A) 36 (B) 6
(C) 12 (D) 24

2. How many subsets does a set with 6 elements have?
(A) 64 (B) 128
(C) 12 (D) 36

3. Using Fermat's little theorem, what is $2^{10} \bmod 11$?
(A) 2 (B) 10
(C) 0 (D) 1

4. The contrapositive of "if p then q" is:
(A) if not p then not q (B) if not q then not p
(C) p and not q (D) if q then p

5. If $|A| = 30$, $|B| = 20$ and $|A \cap B| = 8$, then $|A \cup B|$ is:
(A) 38 (B) 42
(C) 50 (D) 58

6. For $f(x, y) = x^2y + 3y^2$, the partial derivative $\partial f/\partial x$ is:
(A) $2xy + 6y$ (B) $x^2 + 3y^2$
(C) $x^2 + 6y$ (D) $2xy$

7. Which properties of an equivalence relation does "<" on the integers fail?
(A) reflexivity and symmetry (B) transitivity only
(C) reflexivity only (D) none of them

8. A Boolean function of 3 variables has how many minterms in total?
(A) 3 (B) 6
(C) 8 (D) 9

Part B - Long Answer Questions (2 x 10 = 20 Marks)

9. Apply the Euclidean and extended Euclidean algorithms to 1071 and 462. (10 Marks)

(a) State the division algorithm, and state the exact condition under which a has a multiplicative inverse modulo m. *[3]*

(b) Find gcd(1071, 462) by the Euclidean algorithm, then express that gcd in the form 1071x + 462y. *[4]*

(c) Verify your identity by substitution, and state the lcm of the two numbers. *[3]*

10. Explain and apply proof by mathematical induction. (10 Marks)

(a) State the basis step and the inductive step, and explain why a proof that omits either one is worthless. *[4]*

(b) Prove by induction that 3 divides $n^3 - n$ for every integer $n \geq 0$. *[4]*

(c) Give the one-line alternative argument that avoids induction entirely. *[2]*

Unit 2: Vectors and Matrices

Part A - Multiple Choice Questions (8 x 1 = 8 Marks)

1. The determinant of [[3, 8], [4, 6]] is:

(A) −50 (B) −14
(C) 14 (D) 50

2. The dot product (1, 2, 3) · (4, −5, 6) equals:

(A) 32 (B) −12
(C) 4 (D) 12

3. The length of the vector (6, 8) is:

(A) 10 (B) 100
(C) 14 (D) 48

4. Why is a translation in 2-D written as a 3×3 matrix?

(A) so that rotation becomes possible (B) to make the computation faster
(C) because a 2×2 linear map always fixes the origin and cannot add an offset (D) to make the matrix invertible

5. The eigenvalues of [[5, 0], [0, 3]] are:

(A) 0 and 0 (B) 8 and 15
(C) 5 and 3 (D) 4 and 4

6. The first step of Principal Component Analysis is to:

(A) subtract the mean of each feature (B) scale every vector to unit length
(C) compute the eigenvectors (D) sort the features by variance

7. A covariance matrix is always:

(A) diagonal (B) orthogonal
(C) singular (D) symmetric

8. Which interpolation method should be used to resample a segmentation mask?

(A) bicubic (B) nearest neighbour
(C) Gaussian (D) bilinear

Part B - Long Answer Questions (2 x 10 = 20 Marks)

9. Apply Principal Component Analysis by hand to the five points (2, 1), (3, 3), (4, 3), (5, 5) and (6, 8). (10 Marks)

(a) Explain why the data must be centred, and state what the covariance matrix captures. *[3]*

(b) Centre the data and compute the 2×2 covariance matrix. *[4]*

(c) Find both eigenvalues of S and state the percentage of variance carried by the first principal component. *[3]*

10. Apply homogeneous coordinates to rotate the point (4, 2) by 90 degrees anticlockwise about the point (2, 2). (10 Marks)

(a) Explain why the rotation cannot be done with a single 2×2 rotation matrix, and give the sequence of transformations needed. *[3]*

(b) Write the three 3×3 matrices and multiply them into a single matrix M. *[4]*

(c) Apply M to (4, 2) and verify the result geometrically. *[3]*

Unit 3: Probability and Statistics

Part A - Multiple Choice Questions (8 x 1 = 8 Marks)

1. Two fair dice are rolled. What is the probability that the sum is 9?

(A) 4/9 (B) 1/6

(C) 1/12 (D) 1/9

2. The mean of a Binomial(n = 20, p = 0.3) random variable is:

(A) 4.2 (B) 6

(C) 20 (D) 0.3

3. The variance of a Bernoulli(p = 0.4) random variable is:

(A) 0.6 (B) 0.16

(C) 0.24 (D) 0.4

4. For a standard normal distribution, $P(-1 < Z < 1)$ is approximately:

(A) 68 per cent (B) 95 per cent

(C) 99.7 per cent (D) 50 per cent

5. A regression model reports $R^2 = 0.81$. This means:

(A) 81 per cent of the variance in y is explained by the model (B) the slope of the line is 0.81

(C) the correlation is 0.81 (D) the model is 81 per cent accurate

6. A Type I error is:

(A) an error caused by a small sample (B) failing to reject a false null hypothesis

(C) rejecting a true null hypothesis (D) choosing the wrong test statistic

7. If X is uniform on {1, 2, 3, 4, 5, 6}, then E[X] is:

(A) 3 (B) 3.5

(C) 4 (D) 6

8. In Bayes' theorem, the probability assigned before any evidence is seen is called the:

(A) likelihood (B) posterior

(C) evidence (D) prior

Part B - Long Answer Questions (2 x 10 = 20 Marks)

9. Apply Bayes' theorem to a screening test. A disease affects 2 per cent of a population; the test is 95 per cent sensitive and 90 per cent specific. (10 Marks)

(a) State Bayes' theorem, and define prior, likelihood and posterior. *[3]*

(b) Find the probability that a person who tests positive actually has the disease, showing all working. *[4]*

(c) Explain why that figure is so far below 95 per cent. *[3]*

10. Fit and interpret a least-squares regression line for the pairs (1, 2), (2, 4), (3, 5), (4, 4) and (5, 5). (10 Marks)

(a) State what least squares minimises, and explain why the vertical distances are squared rather than simply added. *[3]*

(b) Compute the regression line $y = a + bx$, showing the two sums you use. *[4]*

(c) Compute R^2, interpret it, and state one limitation of predicting y at $x = 6$. *[3]*

Unit 4: Graph Theory

Part A - Multiple Choice Questions (8 x 1 = 8 Marks)

1. A graph has 12 edges. The sum of all its vertex degrees is:

(A) 24 (B) 144

(C) 6 (D) 12

2. How many edges does the complete graph K_6 have?

(A) 6 (B) 12

(C) 15 (D) 30

3. A tree with 10 vertices has how many edges?

(A) 8 (B) 9

(C) 10 (D) 11

4. The chromatic number of any bipartite graph with at least one edge is:

(A) 3 (B) 4

(C) 1 (D) 2

5. Which algorithm can return a wrong answer when some edge weights are negative?

(A) Bellman–Ford (B) Floyd–Warshall

(C) Kruskal (D) Dijkstra

6. A topological sort of a directed graph exists only if the graph is:

(A) bipartite (B) a directed acyclic graph

(C) complete (D) connected

7. The max-flow min-cut theorem states that the maximum flow equals:

(A) the number of augmenting paths used (B) the number of edges in the network

(C) the capacity of the minimum cut (D) the total capacity leaving the source

8. A connected graph has an Eulerian circuit if and only if:

(A) every vertex has even degree (B) it is complete

(C) it contains no cycles (D) exactly two vertices have odd degree

Part B - Long Answer Questions (2 x 10 = 20 Marks)

9. Apply Dijkstra's algorithm to the weighted graph with vertices A, B, C, D, E and edges AB = 4, AC = 2, BC = 1, BD = 5, CD = 8, CE = 10, DE = 2. (10 Marks)

(a) Explain the central idea of Dijkstra's algorithm, and state why it fails on graphs with negative edge weights. *[3]*

(b) Find the shortest distance from A to every other vertex, showing the path in each case. *[4]*

(c) Comment on what this result shows about using the direct edge between two vertices. *[3]*

10. Apply Kruskal's algorithm to find a minimum spanning tree of the same graph (AB = 4, AC = 2, BC = 1, BD = 5, CD = 8, CE = 10, DE = 2). (10 Marks)

(a) State what a minimum spanning tree is, and describe how Kruskal's algorithm builds one. *[3]*

(b) Carry out the algorithm, listing every edge you accept and every edge you reject, and give the total weight. *[4]*

(c) State how many edges an MST on n vertices must contain, and explain why the algorithm may safely reject an edge. *[3]*

Unit 5: Algorithm Complexity Analysis

Part A - Multiple Choice Questions (8 x 1 = 8 Marks)

1. The worst-case running time of binary search on n sorted items is:
(A) O(n log n) (B) O(1)
(C) O(log n) (D) O(n)

2. By the Master theorem, $T(n) = 2T(n/2) + n$ solves to:
(A) $\Theta(n^2)$ (B) $\Theta(n)$
(C) $\Theta(\log n)$ (D) $\Theta(n \log n)$

3. The lower bound on the number of comparisons needed by any comparison-based sorting algorithm is:
(A) $\Omega(n \log n)$ (B) $\Omega(n^2)$
(C) $\Omega(\log n)$ (D) $\Omega(n)$

4. Which of these problems is NP-complete?
(A) binary search (B) 3-SAT
(C) sorting (D) 2-SAT

5. Writing $f(n) = \Theta(g(n))$ means that g is:
(A) a lower bound only (B) both an upper and a lower bound
(C) an upper bound only (D) unrelated to f

6. The recurrence $T(n) = T(n - 1) + 1$ with $T(1) = 1$ solves to:
(A) $\Theta(n)$ (B) $\Theta(n \log n)$
(C) $\Theta(n^2)$ (D) $\Theta(\log n)$

7. The worst-case running time of merge sort is:
(A) $O(n^2)$ (B) O(n)
(C) O(log n) (D) O(n log n)

8. If a problem is in P, then it is:
(A) NP-hard (B) not in NP
(C) also in NP (D) NP-complete

Part B - Long Answer Questions (2 x 10 = 20 Marks)

9. Explain asymptotic notation and apply the Master theorem. (10 Marks)

(a) Define O, Ω and Θ, and state what each one asserts about a running time. *[3]*

(b) Solve $T(n) = 2T(n/2) + n$ and $T(n) = 3T(n/2) + n^2$ using the Master theorem, naming the case used in each. *[4]*

(c) Give one recurrence to which the Master theorem does not apply, and say why. *[3]*

10. Discuss the complexity classes P, NP, NP-hard and NP-complete. (10 Marks)

(a) Define all four classes. *[4]*

(b) Explain the role of a polynomial-time reduction, and state what the Cook–Levin theorem established. *[4]*

(c) State whether 2-SAT is in P and whether 3-SAT is NP-complete, with one sentence on why the difference matters. *[2]*

Unit 6: Algorithmic and Optimization Approaches

Part A - Multiple Choice Questions (8 x 1 = 8 Marks)

1. The fractional knapsack problem is solved optimally by:

(A) divide and conquer
(B) dynamic programming only
(C) backtracking
(D) a greedy choice on the value-to-weight ratio

2. The 0/1 knapsack problem, where items cannot be split, requires:
(A) dynamic programming
(B) a greedy ratio rule
(C) binary search
(D) a minimum spanning tree

3. Dynamic programming applies when a problem has:
(A) overlapping subproblems only
(B) optimal substructure only
(C) both optimal substructure and overlapping subproblems
(D) neither property

4. Huffman coding is an example of which paradigm?
(A) branch and bound
(B) greedy
(C) dynamic programming
(D) backtracking

5. Merge sort is an example of:
(A) dynamic programming
(B) backtracking
(C) divide and conquer
(D) greedy

6. The N-Queens problem is normally solved by:
(A) divide and conquer
(B) backtracking
(C) a shortest-path algorithm
(D) a greedy rule

7. In simulated annealing, as the temperature falls the algorithm:
(A) accepts fewer worsening moves
(B) halts immediately
(C) increases its step size
(D) accepts more worsening moves

8. The No Free Lunch theorem states that:
(A) genetic algorithms outperform all other methods
(B) optimisation problems cannot be solved
(C) a greedy strategy is always optimal
(D) no single algorithm performs best on every problem

Part B - Long Answer Questions (2 x 10 = 20 Marks)

9. Compare the greedy and dynamic-programming approaches on the knapsack problem. Three items have (value, weight) = (60, 10), (100, 20) and (120, 30), and the capacity is 50. (10 Marks)

(a) State the greedy-choice property and optimal substructure, and explain which of the two knapsack versions has each. *[3]*

(b) Solve the fractional version by the greedy ratio rule, and state the true optimum of the 0/1 version. *[4]*

(c) Apply the greedy rule to the 0/1 version, and explain the gap in your answers. *[3]*

10. Explain simulated annealing and the exploration-exploitation trade-off. (10 Marks)

(a) Explain what exploration and exploitation mean, and the role temperature plays in balancing them. *[4]*

(b) The current solution has cost 50 and a candidate move has cost 54, with T = 10. Compute the probability that the move is accepted. *[3]*

(c) State what the algorithm becomes as T approaches 0, and name the risk this creates. *[3]*

----- End of Practice Sheet -----